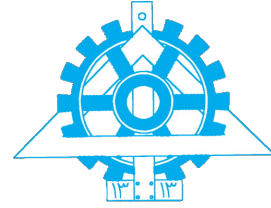




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Statistical Inference

Final Project

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Introduction

This report presents the findings of a comprehensive statistical analysis performed on the Framingham Heart Study dataset. The primary objective was to identify and quantify the risk factors associated with the 10-year risk of coronary heart disease (TenYearCHD). The dataset, consisting of approximately 4,240 participants, includes demographic, behavioral, and medical history variables. By employing data visualization, point estimation, hypothesis testing, and regression modeling, this analysis provides robust statistical evidence linking specific physiological markers to cardiovascular health.

Visualization and Exploratory Data Analysis

This section details the exploratory analysis of the Framingham Heart Study dataset. Visualizations were generated to examine variable distributions, assess the balance of the target variable, and identify potential correlations between physiological markers.

Distributions and Demographics

The distributions of key demographic and health metrics are presented in Figure 1.

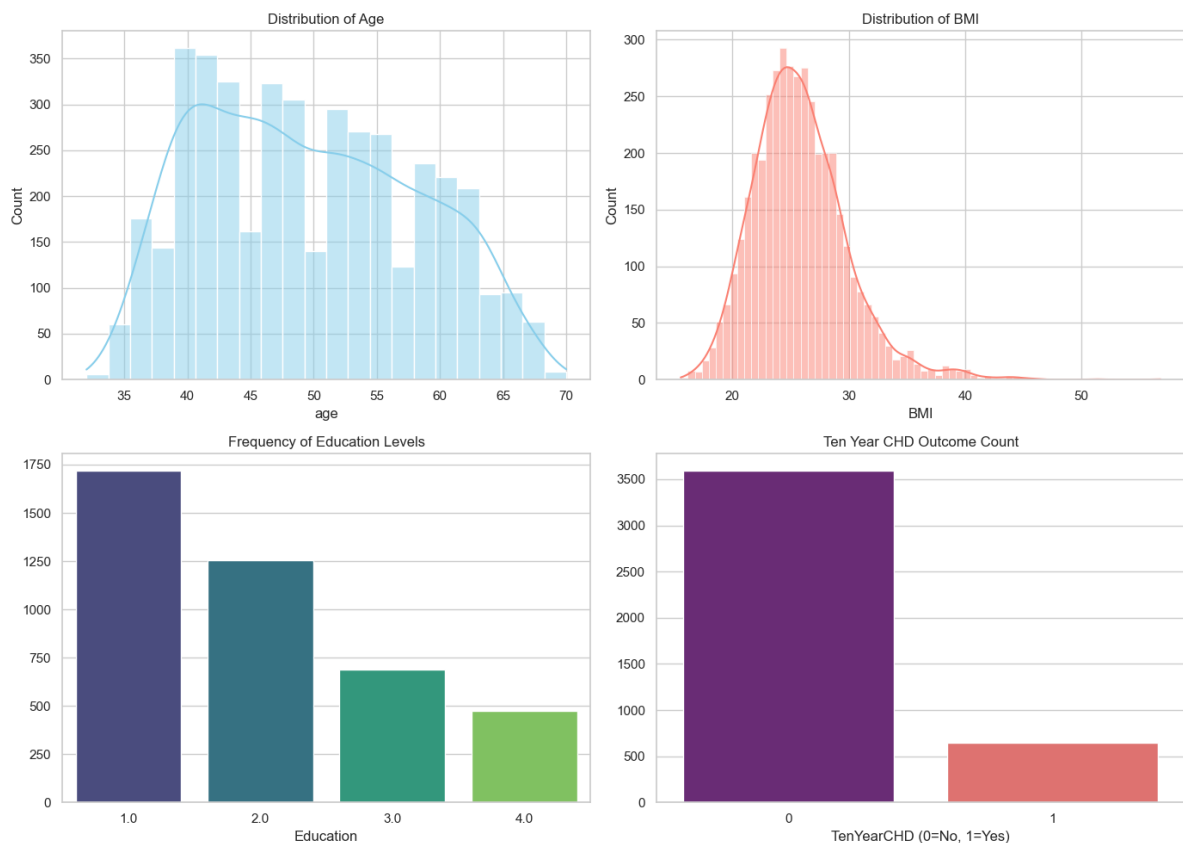


Figure 1: Distributions of Age and BMI (Top), and Frequency of Education and TenYearCHD (Bottom).

The **Age** histogram reveals a broad participation range, with a high concentration of middle-aged individuals, effectively capturing the demographic most at risk for developing chronic cardiovascular conditions.

The **BMI** distribution is roughly normal but exhibits a noticeable right skew. This indicates a concentration of participants in the overweight to obese range ($BMI > 25$), a critical risk factor for heart disease.

The bar charts highlight a significant class imbalance in the target variable, TenYearCHD. The number of participants who did not develop coronary heart disease (0) vastly outnumbers those who did (1). This imbalance suggests that accuracy alone may be a misleading metric for predictive modeling, and sensitivity should be prioritized.

Bivariate Relationships

To understand the interaction between physiological variables, we examined blood pressure and smoking status (Figure 2).

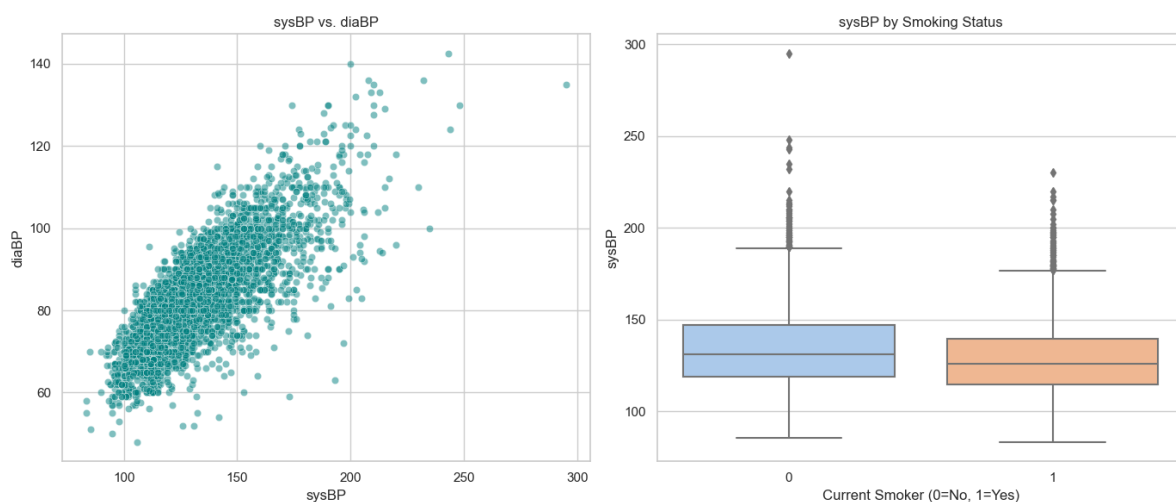


Figure 2: Scatter Plot of Systolic vs. Diastolic BP (Left) and Box Plot of Systolic BP by Smoking Status (Right).

The scatter plot demonstrates a strong, positive linear relationship between **Systolic (sysBP)** and **Diastolic (diaBP)** blood pressure. This collinearity is

physiologically expected but implies that including both variables in a linear regression model could introduce multicollinearity issues.

The box plots comparing sysBP between smokers and non-smokers show similar medians, suggesting that smoking status alone may not drastically shift the central tendency of blood pressure in this cohort. However, the spread and outliers in the smoker group provide a basis for the subsequent T-test to determine if the means are statistically different.

Multivariate Interactions

A pair plot was generated to visualize the interactions between age, BMI, totChol, and glucose, segmented by the target variable TenYearCHD (Figure 3).



Figure 3: Pair Plot of Selected Continuous Variables Colored by TenYearCHD Outcome.

The visualization reveals significant overlap between the two outcome groups across most variables, indicating that no single continuous variable acts as a perfect separator. However, subtle shifts in the distributions of age and glucose for the CHD-positive group are observable, reinforcing their potential as predictors in the logistic regression model.

Correlation Analysis

A heatmap of the correlation matrix (Figure 4) was generated to quantify relationships between all numerical variables.

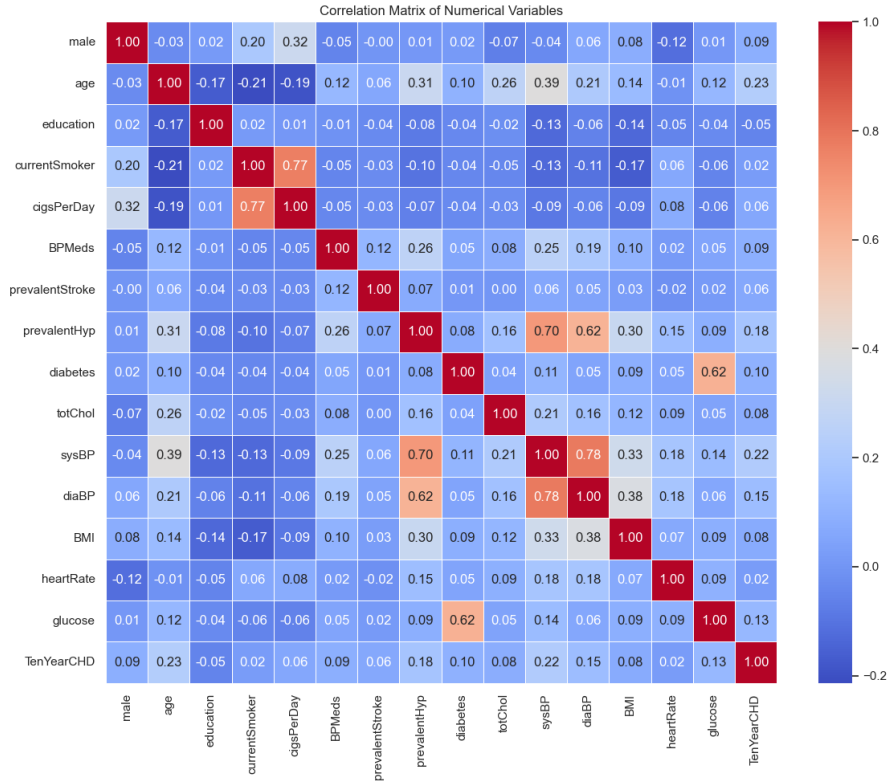


Figure 4: Correlation Matrix of Numerical Variables.

As observed in the scatter plots, sysBP and diaBP show the strongest correlation. Additionally, age exhibits a moderate positive correlation with blood pressure metrics, consistent with arterial stiffening over time.

To investigate the specific relationship between smoking intensity and heart function, a **Spearman's Rank Correlation** was performed between cigsPerDay and heartRate:

- **Spearman's Rho (ρ):** 0.0790
- **P-value:** 2.88×10^{-7}

While the correlation coefficient is weak ($\rho < 0.1$), the p-value is statistically significant ($p < 0.05$). This indicates a positive monotonic relationship: as the number of cigarettes smoked per day increases, heart rate tends to increase, likely due to the sympathomimetic effects of nicotine.

Statistical Estimation

In this phase, we addressed data completeness, estimated population parameters for key cardiovascular metrics, and assessed the distributional properties of the data to validate assumptions for subsequent testing.

Missing Data Handling

Prior to estimation, the dataset was inspected for missing values. Two key variables required imputation to ensure sample size preservation:

- **Glucose:** 388 missing entries were identified. These were imputed using the mean value (≈ 82 mg/dL), assuming the data were missing at random.
- **Education:** 105 missing entries were imputed using the mode (the most frequent education category).

Following this process, all variables used in this analysis contain 0 missing values, ensuring a consistent sample size ($N = 4240$) for the inference procedures.

Point Estimation: Systolic Blood Pressure

We calculated the Maximum Likelihood Estimates (MLE) for the mean and variance of Systolic Blood Pressure (sysBP).

- **MLE for Mean ($\hat{\mu}_{MLE}$):** 132.35 mmHg
- **MLE for Variance ($\hat{\sigma}_{MLE}^2$):** 485.35

It is important to note that the variance estimate uses N in the denominator (rather than $N - 1$), consistent with the definition of the Maximum Likelihood Estimator, which is slightly biased but asymptotically efficient for large samples.

Distribution Fitting: Body Mass Index (BMI)

To determine if parametric tests assuming normality are appropriate for BMI, we performed a Shapiro-Wilk test and visualized the distribution against a theoretical normal curve (Figure 5).

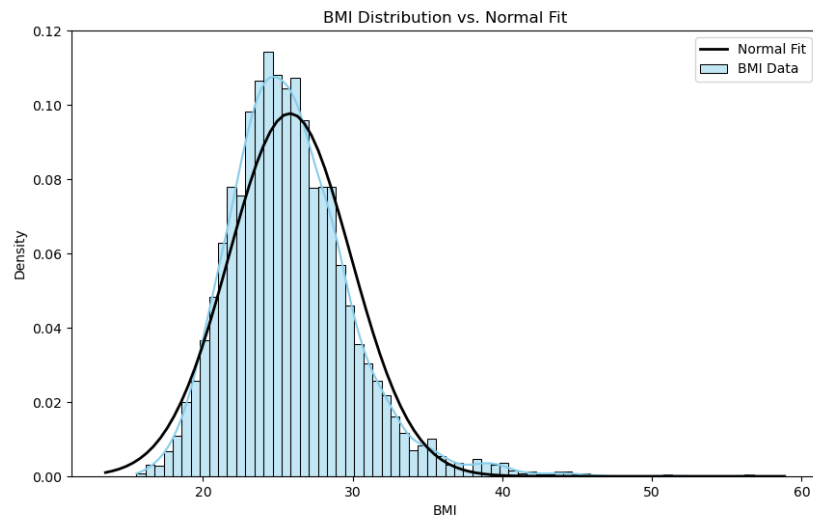


Figure 5: Distribution of BMI with Normal Fit Overlay.

Normality Test Results:

- **Statistic (W):** 0.9576
- **P-value:** 2.05×10^{-33}

Conclusion: Since the p-value is effectively zero ($p < 0.05$), we **reject the null hypothesis** that BMI follows a normal distribution. As seen in Figure 5, the data exhibits a significant right skew (positive skewness), indicating a prevalence of individuals with higher BMI values (overweight/obese) that exceeds what would be expected in a perfectly normal distribution. Consequently, non-parametric methods or robust statistical approaches may be preferred for analyzing BMI.

Interval Estimation: Total Cholesterol

To estimate the true population mean of Total Cholesterol (totChol), we constructed a 95% Confidence Interval (CI). The density plot with the interval boundaries is shown in Figure 6.

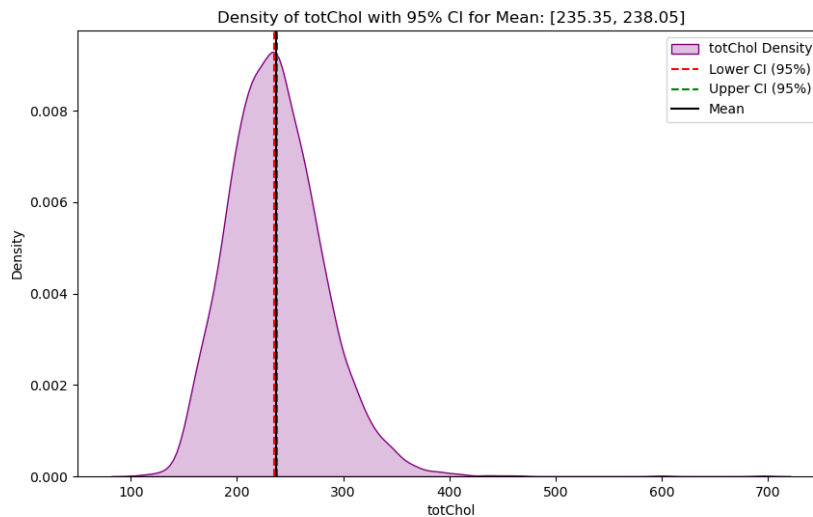


Figure 6: Density of Total Cholesterol with 95% Confidence Interval for the Mean.

Estimation Results:

- **95% Confidence Interval:** (235.35, 238.05) mg/dL.

Interpretation: We are 95% confident that the true average cholesterol level for the Framingham population lies between 235.35 and 238.05 mg/dL. The narrow width of this interval is a direct result of the large sample size ($N \approx 4240$), which significantly reduces the standard error of the mean. This precise estimate establishes a baseline for "average" cholesterol in this cohort, which is clinically elevated compared to modern guidelines (where < 200 mg/dL is desirable).

Hypothesis Testing

This section details the inferential statistics performed to test specific medical hypotheses regarding cardiovascular risk factors.

T-Test: Heart Rate (Smokers vs. Non-Smokers)

Objective: To determine if smoking status is associated with an elevated heart rate.

- **Hypotheses:**

$$H_0 : \mu_{\text{smoker}} \leq \mu_{\text{non-smoker}}$$

$$H_1 : \mu_{\text{smoker}} > \mu_{\text{non-smoker}}$$

- **Test Statistic & P-value:**

$$T = 4.088, \quad p = 2.21 \times 10^{-5}$$

- **Decision:** Reject H_0 .

- **Conclusion:** Since $p < 0.05$, there is statistically significant evidence that smokers have a higher average heart rate than non-smokers in this population. This aligns with the known physiological effect of nicotine as a stimulant.

Power Analysis

Objective: To assess if the sample size was sufficient to detect a medium effect size ($d = 0.5$) with 80% power at a significance level of $\alpha = 0.05$.

- **Result:** The analysis indicated that a sample size of approximately **64 participants per group** (128 total) would be required.
- **Conclusion:** Given our dataset contains over 4,000 records, we have excess statistical power. This ensures that even small, subtle differences between groups (small effect sizes) can be detected with high confidence.

Non-Parametric Test: Glucose (Diabetes Status)

Objective: To compare glucose levels between diabetic and non-diabetic participants. The Mann-Whitney U test was selected because the glucose distribution violated the normality assumption.

- **Hypotheses:**

$$H_0 : P(X_{\text{diabetes}} > Y_{\text{no-diabetes}}) = 0.5 \quad (\text{Distributions are equal})$$

$$H_1 : P(X_{\text{diabetes}} > Y_{\text{no-diabetes}}) \neq 0.5 \quad (\text{Distributions differ})$$

- **Test Statistic & P-value:**

$$U = 353,778, \quad p \approx 2.07 \times 10^{-44}$$

- **Decision:** Reject H_0 .
- **Conclusion:** The p-value is effectively zero, indicating a profound difference in the distribution of glucose levels between the two groups. This statistically validates glucose as a distinct biomarker for diabetes in this dataset.

ANOVA: BMI across Education Levels

Objective: To compare the mean Body Mass Index (BMI) across four levels of education (1: Some High School, 2: High School, 3: Some College, 4: Col-

lege).

- **Hypotheses:**

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$$

H_1 : At least one group mean is different.

- **Test Statistic & P-value:**

$$F = 38.66, \quad p = 1.24 \times 10^{-24}$$

- **Decision:** Reject H_0 .

- **Conclusion:** There is a statistically significant difference in mean BMI across education levels.

Post-hoc Analysis (Tukey's HSD)

To identify which specific groups differed, we performed Tukey's Honestly Significant Difference (HSD) test.

Group 1	Group 2	Mean Diff	p-adj	Lower	Upper	Reject
1.0	2.0	-1.2537	0.0000	-1.6392	-0.8683	True
1.0	3.0	-1.5818	0.0000	-2.0493	-1.1143	True
1.0	4.0	-1.2826	0.0000	-1.8212	-0.7439	True
2.0	3.0	-0.3280	0.3165	-0.8200	0.1639	False
2.0	4.0	-0.0288	0.9992	-0.5888	0.5311	False
3.0	4.0	0.2992	0.6003	-0.3201	0.9185	False

Table 1: Tukey's HSD Results for BMI across Education Levels (FWER=0.05).

Interpretation:

- **Group 1 (Lowest Education) Distinction:** The comparisons involving Group 1 resulted in negative mean differences (e.g., Group 2 mean - Group 1 mean = -1.25). This implies that Group 1 has a significantly **higher** average BMI than all other groups (2, 3, and 4).
- **Homogeneity of Higher Education:** There were no statistically significant differences found between Groups 2, 3, and 4.
- **Key Finding:** Lower educational attainment (specifically "Some High School") is a strong indicator for higher BMI in this cohort, suggesting a socioeconomic link to obesity risk.

Regression Analysis

To quantify the impact of specific risk factors on coronary heart disease and physiological markers, we employed both Logistic Regression (for binary classification) and Lasso Regression (for feature selection).

Logistic Regression: Determinants of CHD Risk

We fitted a logistic regression model to predict the 10-year risk of Coronary Heart Disease (TenYearCHD). All predictors were standardized prior to modeling, allowing for a direct comparison of their relative importance based on the magnitude of their coefficients.

Model Performance:

- **Accuracy:** 85.14%

Interpretation of Standardized Coefficients: The coefficients (β) represent the change in the log-odds of developing CHD for a one-standard-deviation increase in the predictor.

1. **Age ($\beta = 0.5771$):** This is the strongest predictor in the model. As expected, older age significantly increases the probability of developing heart disease.
2. **Systolic Blood Pressure ($\beta = 0.3915$):** Higher systolic pressure is a major, independent risk factor, confirming the link between hypertension and cardiovascular events.
3. **Cigs Per Day ($\beta = 0.2541$):** The quantity of cigarettes smoked is positively associated with risk, distinct from the binary "current smoker" status.

4. **Male** ($\beta = 0.1768$): Being male is associated with a higher baseline risk compared to females.
5. **Glucose** ($\beta = 0.1507$): Elevated glucose contributes to risk, aligning with the diabetic profile.
6. **Total Cholesterol** ($\beta = 0.0184$): Surprisingly, total cholesterol showed a very weak effect in this multivariate model. This suggests that when controlling for age, blood pressure, and glucose, total cholesterol may lose some of its predictive power, or its variance is partially captured by the other metabolic factors.

Lasso Regression: Predicting Systolic Blood Pressure

To identify the most critical determinants of systolic blood pressure (sysBP) and handle potential multicollinearity, we trained a Lasso (L1 Regularization) regression model.

Feature Importance (Magnitude of Coefficients):

- **Diastolic BP** ($\beta = 15.7207$): As expected, diastolic pressure is the dominant predictor of systolic pressure due to their high physiological correlation.
- **Age** ($\beta = 5.0792$): Blood pressure naturally rises with age due to arterial stiffening, making age the second most important determinant.
- **Minor Contributors**: Glucose ($\beta = 1.2259$), Heart Rate ($\beta = 0.9680$), and Total Cholesterol ($\beta = 0.4058$) provided minor positive contributions.
- **CigsPerDay** ($\beta \approx 0$): The Lasso algorithm shrunk the coefficient for cigarettes per day to effectively zero. This suggests that smoking intensity is not a strong *direct* linear predictor of systolic blood pressure when diastolic pressure is already included in the model.

Model Diagnostics

To ensure the reliability of our regression estimates, we performed diagnostic checks for homoscedasticity and multicollinearity.

Homoscedasticity Check

Figure 7 displays the residuals versus fitted values for the linear model predicting sysBP.

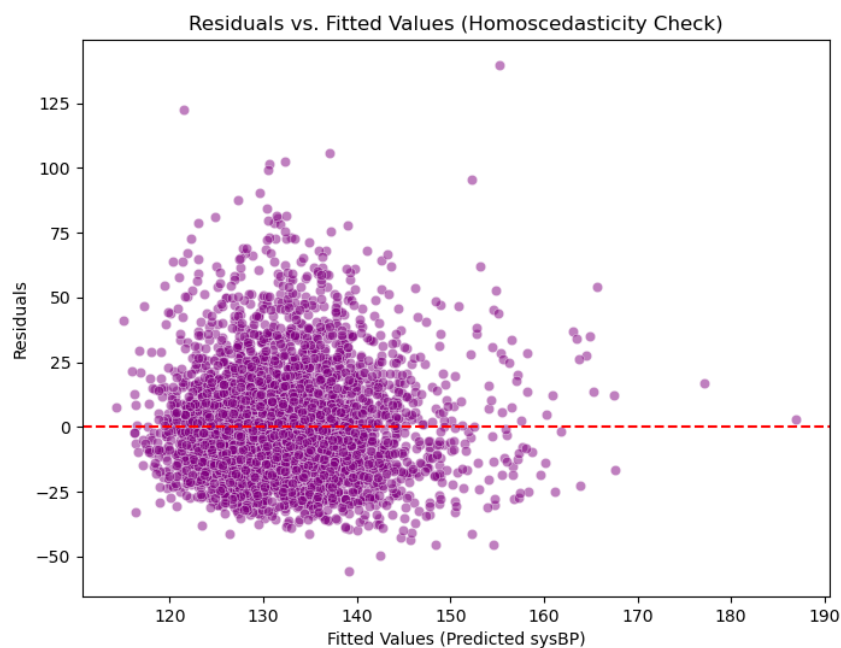


Figure 7: Residuals vs. Fitted Values for sysBP Regression.

Ideally, residuals should be randomly scattered around zero. A "funnel" shape would indicate heteroscedasticity (where variance changes with the fitted value). The observed scatter suggests the linear assumption holds reasonably well, though some variance at higher blood pressure levels is typical in medical data.

Multicollinearity (VIF)

We calculated the Variance Inflation Factor (VIF) to detect multicollinearity among the predictors. A VIF value exceeding 5 (or strictly 2.5) indicates problematic correlation.

Feature	VIF
Age	1.16
CigsPerDay	1.05
TotChol	1.10
DiaBP	1.24
BMI	1.18
HeartRate	1.06
Glucose	1.03

Table 2: Variance Inflation Factors (VIF) for Predictors.

Conclusion: All VIF values are below 1.3. This indicates that there is **no significant multicollinearity** among these predictors. The coefficients in our regression models are therefore stable, and the standard errors are not artificially inflated, ensuring reliable inference.

Conclusion

This statistical inference project successfully utilized the Framingham Heart Study dataset to isolate key drivers of coronary heart disease.

1. **Primary Risk Factors:** Age and Systolic Blood Pressure are the most dominant predictors of 10-year CHD risk.
2. **Socioeconomic Link:** Lower education is statistically linked to higher BMI, suggesting public health interventions should target lower-income demographics.
3. **Behavioral Impact:** Smoking is significantly associated with elevated heart rate and increased CHD risk, distinct from its effect on blood pressure.

The findings validate established medical knowledge using rigorous statistical methods, demonstrating the power of large-scale cohort studies in epidemiological inference.

References

Aasheesh. (n.d.). *Framingham Heart Study Dataset*. Kaggle. Retrieved February 12, 2026, from <https://www.kaggle.com/datasets/aasheesh200/framingham-heart-study-dataset/data>